# Reproducibility

## Reproducibility I

Reproducibility in computational research has well-documented prerequisites: open data, open code, and a deterministic build that can be re-run from scratch (Peng 2011). The bundled `manuscript/config.yaml` is intentionally configured to satisfy all three for **strict reproducibility**:

# Reproducibility I

1. `project_config.search.sources: [local]` consumes `data/corpus.json`, which is a curated and committed JSON corpus. No network is required to run the pipeline.
2. `search.cache_dir: output/search/cache` writes deterministic JSON cache files; running the same query twice produces a byte-identical artifact tree (modulo timestamp metadata in the cache file itself).
3. `enrichment.fetch_abstracts: true` reads abstracts directly from the corpus when present; no network fetch is required.
4. `enrichment.fetch_fulltext: false` is the default — full-text fetching is opt-in and gated behind the optional `pypdf` dependency (`uv sync --group rendering`).
5. `llm.enabled: false` is the default — the LLM stage is opt-in and requires a running `ollama serve`. When enabled, `seed: 42` and `temperature: 0.0` are pinned.

## Switching to live search I

Replace the `sources` list with the desired backend set:

```
search:
  query: "your topic"
  sources: [arxiv, crossref]
  crossref_mailto: "you@example.org"
```

## Switching to live search I

A first live run populates `output/search/cache/`. Commit that directory to the repo and the pipeline becomes reproducible across machines without further configuration changes.

## Determinism guarantees I

The full determinism contract is itemised in tbl. 1: every pipeline stage is annotated as fully, conditionally, or non-deterministic, with an explicit mechanism column so reviewers can audit each row independently.

# Determinism guarantees I

Table 1: Determinism contract by pipeline stage. Cached stages are byte-stable across reruns; live stages depend on the upstream source and are pinned to the cache file once a successful run completes.

| Stage | Deterministic? | Mechanism |
|---|---|---|
| Search (cached hit) | yes | `SearchCache` JSON files |
| Search (cache miss) | no | live API |
| Dedup / merge | yes | DOI / arXiv-id canonical keys; tie-break by score then year |
| Citation-key generation | yes | unicode folding + stop-word skip; collision suffix is deterministic |
| BibTeX writer | yes | byte-stable format pinning (verified by `tests/infra_tests/reference/`) |
| Abstract fetch | yes (cached) / no (live) | per-paper `<safe_id>.txt` cache |
| Fulltext fetch | yes (cached) / mostly (live) | per-paper `<safe_id>.{pdf,txt}` cache; live fetch's `pypdf` text extraction is not bit-stable across versions |
| LLM synthesis | mostly | `seed=42`, `temperature=0.0`; Ollama deterministic up to its own minor variance |

# Determinism guarantees I

| Stage | Deterministic? | Mechanism |
|---|---|---|
| Figure generation | yes (within Matplotlib version) | fixed palette, fixed bin width, no random subsampling |

## Verifying reproducibility locally I

```
# Run twice; nothing in output/ should diff except
the cache timestamps.
uv run python projects/templates/template_search_proj
ect/scripts/run_search_pipeline.py
mv projects/templates/template_search_project/output
projects/templates/template_search_project/output_fir
st
uv run python projects/templates/template_search_proj
ect/scripts/run_search_pipeline.py
diff -ru \
    projects/templates/template_search_project/output
_first/corpus.json \
    projects/templates/template_search_project/output
/corpus.json
```

## Verifying reproducibility locally I

The only expected differences are inside `output/search/cache/search_*.json`, where `_cached_at` is wall-clock time at write.

## Limitations I

The reproducibility contract enumerated in sec. `sec:reproducibility` (items 1–5 and tbl. 1) does not eliminate the following well-defined sources of non-reproducibility, which are surfaced here so reviewers can audit them explicitly rather than inferring from the contract table:

## Limitations I

- **Live search drift.** When `project_config.search.sources` includes `arxiv` or `crossref`, the first cache-miss invocation hits the live API; the cached JSON freezes that response, but two cold-start clones running on different days will see different paper sets. Pin a `LocalBackend` corpus or commit `output/search/cache/` to break this dependency.
- **`pypdf` version drift.** The fulltext fetcher uses `pypdf` to extract text from a downloaded PDF. `pypdf`'s text-extraction algorithm is not bit-stable across major versions; upgrading `pypdf` can produce different `<safe_id>.txt` cache contents from the same source PDF. The PDF bytes themselves are bit-stable so the cache freezes the inputs, not the extraction.
- **Ollama version drift.** Pinning `seed=42` and `temperature=0.0` controls Ollama's sampling, but the model weights, tokenizer, and template can change between Ollama releases. Document the Ollama version alongside `config.llm.model` when archiving a run for replication.
- **Paperclip backend status.** The `paperclip` backend is opt-in and the production endpoint may reject the adapter's request method; the run records the response in `SearchResult.errors[paperclip]` and continues. Treat `paperclip` results as advisory until the upstream service stabilises.

# Limitations II

- **External backend behaviour outside this project's control.** arXiv and Crossref are the source of truth; this project is faithful to whatever they return. A retraction, metadata fix, or DOI assignment upstream will alter the cache on the next cold-start invocation.

## Limitations I

These limitations bound *what* the cache + seed + corpus pinning achieves. Inside those bounds, the contract in tbl. 1 is total: every cached pipeline stage is byte-stable across reruns (verified by `tests/test_pipeline.py::TestRunLiteraturePipeline::test_bibtex_byte_identical_across_reruns`).